

OLTENITA, Romania

WMO: 154750

Lat: 44.0747N

Lon: 26.6372E

Elev: 52

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	16.2	8.2	8.4	12.4	13.1	10.7	18.5	24.2	21.6	21.8	30.5	4.1	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.4	92.9	72.6	90.5	71.5	88.2	70.5	75.7	88.2	74.3	86.0	72.9	83.9	5.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	119.7	80.1	70.8	113.7	78.9	69.3	108.1	77.5	39.1	88.1	37.8	85.8	36.5	84.2	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.9	16.6	13.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	29.9	38.9	45.5	53.1	63.6	71.9	74.5	74.7	66.5	53.3	43.2	35.9
	DBStd	16.72	9.09	8.82	8.40	7.42	5.84	5.09	4.37	4.74	6.64	7.58	8.18	6.85
	HDD50	1914	623	321	186	54	1	0	0	0	0	55	235	439
	HDD65	4876	1088	732	605	359	95	8	0	2	64	367	654	902
	CDD50	3492	0	9	46	147	424	656	760	765	494	158	31	2
	CDD65	979	0	0	0	2	53	214	296	302	107	5	0	0
	CDH74	9915	0	0	8	101	565	1956	2783	3308	1110	81	3	0
	CDH80	3932	0	0	0	16	121	738	1132	1524	386	15	0	0
Wind	WSAvg	5.1	5.5	5.6	5.7	5.8	5.7	4.8	4.5	4.4	4.5	4.5	4.6	5.8
Precipitation	PrecAvg	23.4	1.7	1.1	1.6	1.9	2.4	2.6	2.4	1.5	2.2	2.0	1.7	2.0
	PrecMax	31.5	4.5	3.7	3.9	5.0	8.2	6.8	5.7	4.8	6.5	5.0	3.7	5.8
	PrecMin	14.3	0.3	0.0	0.1	0.1	0.5	0.7	0.2	0.1	0.0	0.1	0.0	0.0
	PrecStd	5.0	1.1	0.8	0.9	1.0	1.7	1.4	1.4	1.2	1.6	1.3	1.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	68.5	73.8	82.4	88.4	95.1	96.0	96.0	91.7	83.2	72.4	59.0
		MCWB	48.2	57.5	60.0	63.7	71.6	75.5	76.1	73.5	69.4	65.8	60.3	52.1
	2%	DB	49.7	60.3	68.9	76.5	83.5	90.5	92.0	93.0	88.2	75.9	65.2	53.7
		MCWB	45.0	51.7	56.1	61.2	67.3	73.0	72.7	72.1	68.1	62.3	57.6	48.0
	5%	DB	45.3	55.9	64.4	71.8	80.5	87.4	89.3	90.7	84.5	71.3	60.2	49.1
		MCWB	42.2	48.9	53.7	58.4	65.8	70.9	71.9	71.1	66.8	60.3	54.7	45.1
	10%	DB	41.2	51.7	59.6	67.9	76.9	84.3	86.2	88.3	80.8	66.9	55.4	44.9
		MCWB	39.5	46.4	51.4	56.6	64.2	70.4	70.2	70.5	66.2	58.2	52.2	42.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	57.3	63.1	64.9	73.0	79.0	78.1	77.1	74.6	67.3	61.1	53.1
		MCDB	54.3	67.4	72.4	79.5	83.8	92.1	93.2	89.5	86.9	78.3	70.5	59.1
	2%	WB	45.5	52.6	57.6	62.5	69.7	75.1	75.3	75.5	71.5	63.6	58.4	48.8
		MCDB	49.2	59.2	66.2	75.0	80.0	87.0	88.1	87.4	81.5	73.1	65.4	53.2
	5%	WB	42.4	49.6	54.6	60.3	67.4	73.2	73.7	74.0	69.3	61.3	55.6	45.8
		MCDB	44.6	54.9	63.1	70.9	77.0	84.5	85.7	85.3	78.8	69.7	59.7	49.1
	10%	WB	39.7	47.0	51.9	57.9	65.5	71.6	72.1	72.6	67.5	59.3	52.9	42.8
		MCDB	41.2	51.5	59.3	66.0	74.7	82.3	82.7	83.4	77.0	65.8	55.9	44.9
Mean Daily Temperature Range	5% DB	MDBR	12.2	15.8	18.8	21.0	20.6	20.6	21.1	24.4	22.9	20.2	12.8	11.4
		MCDBR	17.7	24.5	28.1	30.1	26.2	25.9	25.8	28.1	29.3	29.2	21.6	19.0
		MCWBR	13.2	15.8	16.3	16.1	12.1	10.6	10.4	10.5	11.7	16.0	12.8	13.6
	5% WB	MCDBR	16.5	22.4	27.5	28.3	22.4	24.4	22.6	24.1	24.6	26.2	18.9	18.9
		MCWBR	12.8	15.2	16.4	15.4	11.4	11.1	10.1	10.0	10.9	15.0	11.6	13.6
Clear-Sky Solar Irradiance	taub	0.352	0.385	0.421	0.462	0.435	0.443	0.453	0.467	0.434	0.405	0.377	0.355	
	taud	2.262	2.185	2.141	2.047	2.203	2.215	2.196	2.151	2.207	2.264	2.291	2.276	
	Ebn at Noon	236	248	255	255	266	264	259	251	248	239	225	221	
	Edh at Noon	28	36	43	51	45	44	45	45	39	33	27	25	
All-Sky Solar Radiation	RadAvg	449	703	1074	1485	1859	2017	2049	1836	1334	862	488	374	
	RadStd	69	102	108	159	138	144	137	98	103	96	65	50	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (55 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-214	+126	+74	

Nomenclature: See separate page